UNIT 1: THEORY OF MATRICES

1. Find the rank of the Matrices using Echelon form — Solutions

I. $\begin{bmatrix} 4 & 2 & -1 & 2 \\ 1 & -1 & 2 & 1 \\ 2 & 2 & -2 & 0 \end{bmatrix}$ — 2

II. $\begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$ — 2

III. $\begin{bmatrix} 2 & -2 & 0 & 6 \\ 4 & 2 & 0 & 2 \\ 1 & -1 & 0 & 3 \\ 1 & -2 & 1 & 2 \end{bmatrix}$ — 3

2. Find the rank of the Matrices using normal form

I. $\begin{bmatrix} 1 & -1 & 2 & 3 \\ 4 & 1 & 0 & 2 \\ 0 & 3 & 1 & 4 \\ 0 & 1 & 0 & 2 \end{bmatrix}$ — 4

II. $\begin{bmatrix} 1 & 2 & 1 & 0 \\ -2 & 4 & 3 & 0 \\ 1 & 0 & 2 & -8 \end{bmatrix}$ — 3

III. $\begin{bmatrix} 6 & 1 & 3 & 8 \\ 4 & 2 & 6 & -1 \\ 10 & 3 & 9 & 7 \\ 16 & 4 & 12 & 15 \end{bmatrix}$ — 2

3. Examine for consistency and if consistent then solve it.

I. $\begin{cases} 4x - 2y + 6z = 8 \\ x + y - 3z = -1 \\ 15x - 3y + 9z = 21 \end{cases}$ $\begin{cases} x = 1 \\ y = 3t - 2 \\ z = t \end{cases}$

II. $\begin{cases} 2x + z = 4 \\ x - 2y + 2z = 7 \\ 3x + 2y = 1 \end{cases}$ $\begin{cases} x = 2 - \frac{t}{2} \\ y = -\frac{5}{2} + \frac{3t}{4} \\ z = t \end{cases}$

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III. $\begin{cases} 2x_1 + x_2 - 5x_3 + x_4 = 8 \\ x_1 + 3x_2 - 6x_4 = -15 \\ 2x_2 - x_3 + 2x_4 = -5 \\ x_1 + 4x_2 - 7x_3 + 6x_4 = 0 \end{cases}$ $\begin{cases} x_1 = 3 \\ x_2 = -4 \\ x_3 = -1 \\ x_4 = 1 \end{cases}$

IV. $\begin{cases} x_1 + x_2 + 2x_3 + x_4 = 5 \\ 2x_1 + 3x_2 - x_3 - 2x_4 = 2 \\ 4x_1 + 5x_2 + 3x_3 = 7 \end{cases}$ Inconsistent

V. $\begin{cases} x + 2y + 3z = 0 \\ 2x + 3y + z = 0 \\ 4x + 5y + 4z = 0 \end{cases}$ $\begin{cases} x = 0 \\ y = 0 \\ z = 0 \end{cases}$

VI. $\begin{cases} 2x - y + 3z = 0 \\ 3x + 2y + z = 0 \\ x - 4y + 5z = 0 \end{cases}$ $\begin{cases} x = t \\ y = -t \\ z = t \end{cases}$

4. Investigate for what values of a & b, the system of simultaneous equation

$$2x - y + 3z = 2$$

$$x + y + 2z = 2$$

$$5x - y + az = b$$

Have (1) No solution (2) A unique solution (3) An infinite number of solutions.

(**Solutions** (1) $a = 8, b \neq 6$ (2) $a \neq 8, b \in R$ (3) $a = 8, b = 6$)

5. Investigate for what values of k the equations

$$x + y + z = 1$$

$$2x + y + 4z = k$$

$$4x + y + 10z = k^2$$

Have infinite number of solutions? Hence, find solutions.

(**Solutions** : $k = 1,2$)

6. Examine for Linear dependence or independence the following system of vectors. If dependent, find the relation between them

I. $x_1 = (1, -1, 1),\ x_2 = (2,1,1),\ x_3 = (3,0,2)$ Dependent $x_1 + x_2 = x_3$

II. $x_1 = (1,1,1,3),\ x_2 = (1,2,3,4),\ x_3 = (2,3,4,7)$ Dependent $x_1 + x_2 = x_3$

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III. $x_1 = (3,1,-4),\ x_2 = (2,2,-3),\ x_3 = (0,-4,1)$ Dependent

$2x_1 = 3x_2 + x_3$

IV. $x_1 = \begin{pmatrix}1\\2\\3\end{pmatrix}, x_2 = \begin{pmatrix}3\\-2\\1\end{pmatrix}, x_3 = \begin{pmatrix}1\\-6\\-5\end{pmatrix}$ Dependent

$2x_1 + x_3 = x_2$

7. Given the transformation $= \begin{bmatrix}1 & -2 & 3\\2 & 0 & -3\\1 & 1 & 1\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}$. Find the co-ordinates (x_1, x_2, x_3) corresponding to $(2,3,0)$ in Y.

$x_1 = \frac{21}{19}$

$x_2 = -\frac{16}{19}$

$x_3 = -\frac{5}{19}$

8. Express each of the transformation $x_1 = 3y_1 + 5y_2$ and $x_2 = -y_1 + 7y_2$

$y_1 = z_1 + 3z_2$

$y_2 = 4z_1$

In the matrix form and find the composite transformation which expresses x_1, x_2 in terms of z_1, z_2.

$x_1 = 23z_1 + 9z_2$

$x_2 = 27z_1 - 3z_2$

9. Verify whether the following matrices are orthogonal or not, if so write A^{-1} :

I. $A = \begin{bmatrix}\frac{1}{\sqrt{3}} & 0 & \frac{2}{\sqrt{6}}\\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}}\\ -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}}\end{bmatrix}$ Yes

II. $A = \frac{1}{3}\begin{bmatrix}2 & 2 & 1\\-2 & 1 & 2\\1 & -1 & 2\end{bmatrix}$ No

10. If $A = \begin{bmatrix}\frac{1}{3} & \frac{2}{3} & a\\ \frac{2}{3} & \frac{1}{3} & b\\ \frac{2}{3} & -\frac{2}{3} & c\end{bmatrix}$ is orthogonal, Find a,b,c.

$a = \pm\frac{2}{3}$

$b = \mp\frac{2}{3}$

$c = \pm\frac{1}{3}$

11. Find the Eigen values and corresponding Eigen vectors for the following matrices

I. $A = \begin{bmatrix}9 & -1 & 9\\3 & -1 & 3\\-7 & 1 & -7\end{bmatrix}$

(**Solution** :- $-1,0,2$ *and* $\begin{bmatrix}1\\1\\-1\end{bmatrix}, \begin{bmatrix}1\\0\\-1\end{bmatrix}, \begin{bmatrix}4\\1\\-3\end{bmatrix}$)

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II. $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & -1 & 2 \\ 0 & 1 & 1 \end{bmatrix}$

(**Solution** :- $0, 2, -2$ *and* $\begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -3 \\ 1 \end{bmatrix}$)

III. $A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$

(**Solution** :- $5, -3, -3$ *and* $\begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}$)

IV. $A = \begin{bmatrix} -3 & -7 & -5 \\ 2 & 4 & 3 \\ 1 & 2 & 2 \end{bmatrix}$

(**Solution** :- $1,1,1$ *and* $\begin{bmatrix} -3 \\ 1 \\ 1 \end{bmatrix}$)

V. $A = \begin{bmatrix} 3 & 10 & 5 \\ -2 & -3 & -4 \\ 3 & 5 & 7 \end{bmatrix}$

(**Solution** :- $3,2,2$ *and* $\begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}, \begin{bmatrix} 5 \\ 2 \\ -5 \end{bmatrix}$)

12. Verify Cayley-Hamilton theorem for the following matrix and use it find Inverse:

I. $A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$

II. $A = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 2 & 4 \\ 0 & 0 & 2 \end{bmatrix}$

13. Find A^4 with the help of Cayley Hamilton theorem

If $A = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$.

(**Solution** :- $\begin{bmatrix} -49 & -50 & -40 \\ 65 & 66 & 40 \\ 130 & 130 & 81 \end{bmatrix}$)

14. If $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$, then express $A^5 - 4A^4 - 7A^3 + 11A^2 - A - 10I$ in terms of A.

(**Solution** :- $A + 5I$)

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15. Consider the triangle with Vertices $A(1,4), B(5,3)$ *and* $C(1,1)$ then

I. Rotate the triangle 90^0 clockwise.
(Solution: A'(1,-1),B'(3,-5),C'(4,-1))

II. Rotate the triangle 90^0 counter clockwise.
(Solution: A'(-1,1),B'(-3,5),C'(-4,1))

III. Take the reflection about X-axis
(Solution: A'(1,-1),B'(5,-3),C'(1,-4))

IV. Take its reflection about Y- axis
(Solution: A'(-1,1),B'(-5,3),C'(-1,4))

V. Translate the triangle 6 units right and 5 units down
(Solution: A'(7,-4),B'(11,-2),C'(7,-1))

16. Centre of the arc of the circle in a given coordinate system is $(100,100,100)$. Origin is shifted to the point $(-10,-5,-2)$.Rotation is carried out about Y axis through an ang of 30^0. Find the centre of the arc of the circle in new coordinate system. (46.66,105,134.66)